

Database tables

```
mysql> describe online_users;
```

Field	Type	Null	Key	Default	Extra
username	varchar(50)	NO	PRI	NULL	

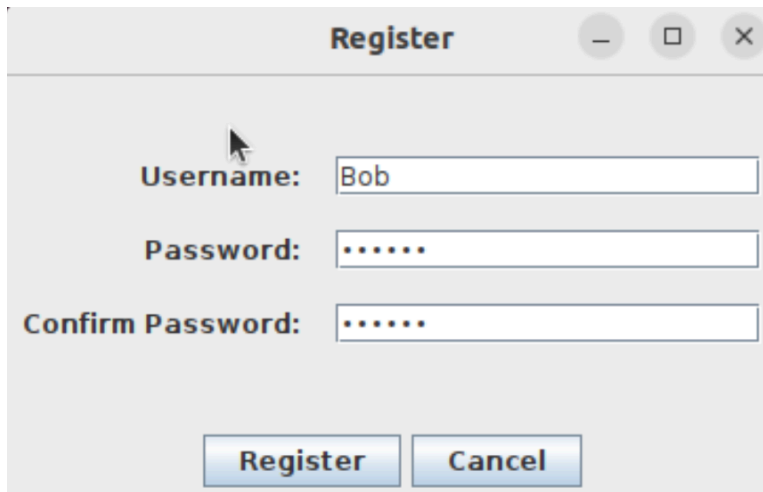
1 row in set (0.00 sec)

```
mysql> describe users;
```

Field	Type	Null	Key	Default	Extra
username	varchar(50)	NO	PRI	NULL	
password	varchar(255)	NO		NULL	
num_games	int	NO		0	
num_wins	int	NO		0	
avg_time_to_win	float	NO		0	
leaderboard_rank	int	NO		0	

6 rows in set (0.00 sec)

Register



The image shows a 'Register' window with a title bar containing standard window controls. Inside, there are three input fields: 'Username:' with the text 'Bob', 'Password:' with six dots, and 'Confirm Password:' with six dots. At the bottom, there are two buttons: 'Register' and 'Cancel'.

When register is pressed, the following steps are executed:

1. Check whether username already exists by querying users table. (READ) - This step is redundant, it is only here to provide clearer error messages for users.
2. If it doesn't, insert a new entry in users table. (INSERT)

Registering with 'Bob' username

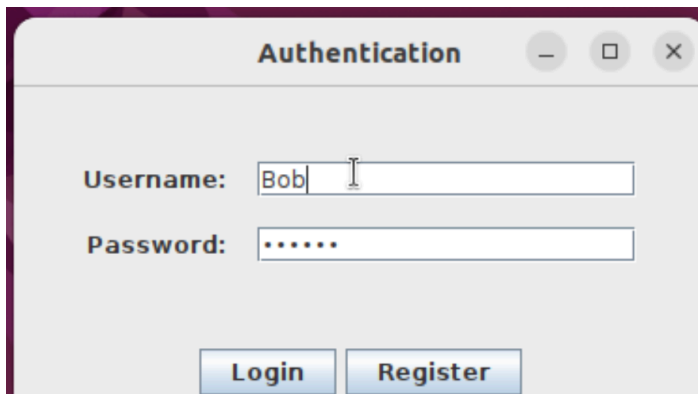
Before

```
mysql> select * from users;
+-----+-----+-----+-----+-----+-----+
| username | password | num_games | num_wins | avg_time_to_win | leaderboard_rank |
+-----+-----+-----+-----+-----+-----+
| a        | a        | 5         | 0        | 0               | 0                |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

After

```
mysql> select * from users;
+-----+-----+-----+-----+-----+-----+
| username | password | num_games | num_wins | avg_time_to_win | leaderboard_rank |
+-----+-----+-----+-----+-----+-----+
| a        | a        | 5         | 0        | 0               | 0                |
| Bob      | 123456   | 0         | 0        | 0               | 0                |
+-----+-----+-----+-----+-----+-----+
```

Login



The image shows a window titled "Authentication" with a light gray background. It contains two input fields: "Username:" with the text "Bob" and a cursor at the end, and "Password:" with six dots. Below the fields are two buttons: "Login" and "Register".

1. Check whether username and password are correct by querying users table (READ)
2. If correct, check whether user is already logged in by querying online_users table (READ) - This step is redundant, it is only here to provide clearer error messages for users.
3. If not logged in already, insert username into online_users table. (INSERT)

Logging in with 'Bob' username

Before

```
mysql> select * from online_users;
+-----+
| username |
+-----+
| a        |
+-----+
1 row in set (0.00 sec)
```

After

```
mysql> select * from online_users;
+-----+
| username |
+-----+
| a        |
| Bob      |
+-----+
2 rows in set (0.00 sec)
```

Logout



1. Remove username from online_users (DELETE)

Logging out as 'Bob' user

Before

```
mysql> select * from online_users;
+-----+
| username |
+-----+
| a        |
| Bob      |
+-----+
2 rows in set (0.00 sec)
```

After

```
mysql> select * from online_users;
+-----+
| username |
+-----+
| a        |
+-----+
1 row in set (0.00 sec)
```

Fetching user info (User Profile + Leaderboard)

User profile section

1. Query users table to fetch information for currently logged in user (READ)

```
mysql> select * from users where username = "Bob";
+-----+-----+-----+-----+-----+-----+
| username | password | num_games | num_wins | avg_time_to_win | leaderboard_rank |
+-----+-----+-----+-----+-----+-----+
| Bob      | 123456   | 1         | 0        | 0               | 0                |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

User Profile

Play Game

Leaderboard

Logout

Name:

Bob

Number of Wins:

0

Number of Games:

1

Average Time to Win:

0.0

Rank:

0

Leaderboard section

1. Query users table to fetch information for all users. (READ)

```
mysql> select * from users;
+-----+-----+-----+-----+-----+-----+
| username | password | num_games | num_wins | avg_time_to_win | leaderboard_rank |
+-----+-----+-----+-----+-----+-----+
| a        | a        | 5         | 0        | 0               | 0                |
| Bob      | 123456   | 1         | 0        | 0               | 0                |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Game Dashboard				
User Profile	Play Game	Leaderboard	Logout	
Rank	Name	Wins	Games	Avg Time to Win
0	a	0	5	0.0
0	Bob	0	1	0.0

Play game



For the sake of demonstrating the UPDATE operation, the 'Play Game' button will increment users' num_games by 1 when clicked.

Pressing "Play Game" as Bob

Before

```
mysql> select num_games from users where username = "Bob";
+-----+
| num_games |
+-----+
|          1 |
+-----+
1 row in set (0.00 sec)
```

After

```
mysql> select num_games from users where username = "Bob";
+-----+
| num_games |
+-----+
|          2 |
+-----+
1 row in set (0.00 sec)
```

Update password

Also for the sake of demonstrating the UPDATE operation, there is a button in 'User Profile' to change the current user's password.

Name:	ric
Number of Wins:	0
Number of Games:	0
Average Time to Win:	0.0
Rank:	0
Update Password	

Update Password

×

Current password:

.....

New password:

.....

Confirm new password:

.....

OK

Cancel

Pressing 'Update Password' as Bob

Before

```
mysql> select * from users where username = 'bob';
+-----+-----+-----+-----+-----+-----+
| username | password | num_games | num_wins | avg_time_to_win | leaderboard_rank |
+-----+-----+-----+-----+-----+-----+
| Bob      | 123456   | 2         | 0        | 0               | 0                |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.01 sec)
```

After

```
mysql> select * from users where username = 'bob';
+-----+-----+-----+-----+-----+-----+
| username | password | num_games | num_wins | avg_time_to_win | leaderboard_rank |
+-----+-----+-----+-----+-----+-----+
| Bob      | newpass  | 2         | 0        | 0               | 0                |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Consider whether transactions should be implemented in this application, and provide your reasoning.

As the application currently stands, there is no need for transactions to be implemented. The only scenario where it could be considered is during registration and login where we are checking specific conditions before inserting new entries (to either users or online_users). However, even if race conditions occur after the respective checks, the primary key constraint (username) will prevent entries with duplicate usernames from being inserted, thus the integrity of the database will still be maintained.

Consider the example for login: (steps will follow  dist_assg2_part2)

1. User1 passes step 1
2. User2 passes step 1
3. User1 passes step 2
4. User2 passes step 2
5. User1 completes step 3 with username "Bob".
6. User2 attempts step 3 with username "Bob". The server will receive an error from the database, reminding that duplicate primary keys are prohibited.

A similar process happens for register when dealing with this type of race condition.